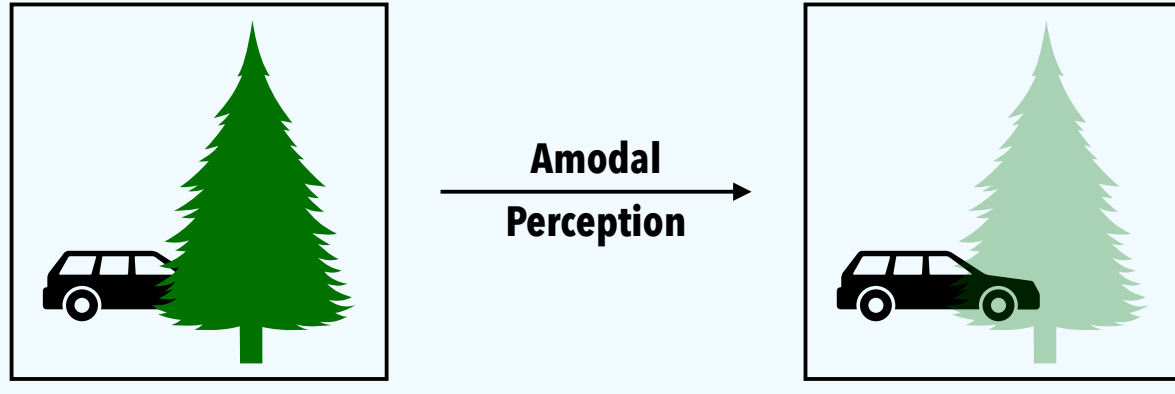


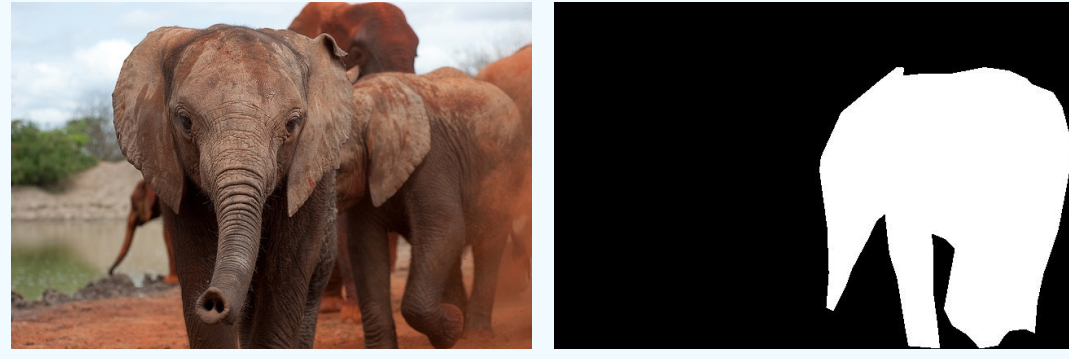
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Introduction

- Amodal perception is to **infer the complete shape of occluded objects**. Amodal perception is a **vital** ability of human's cognitive system and shows essential potential for **tremendous** real-world applications including autonomous driving, robotic gripping and novel view synthesis.



- In computer vision, amodal instance segmentation has aroused **broad** concern since it was proposed, which aims to predict complete shapes of partially occluded objects.
- However, annotating pixel-level ground-truth amodal masks for such objects is **labor-intensive and error-prone** due to the absence of visible cues in occluded regions.

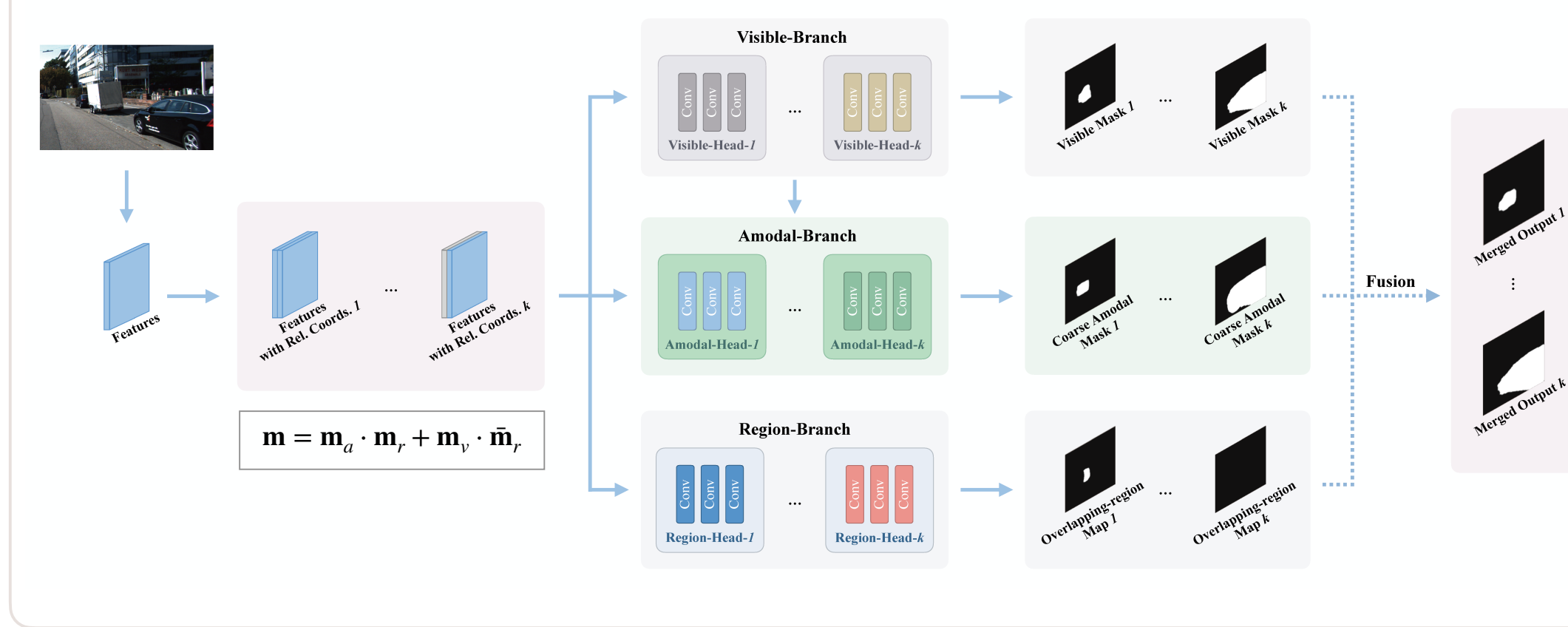


- To solve the challenges of pixel-level annotation, Bayesian-Amodal (Sun et al., 2022), a **weakly supervised** approach is proposed that utilizes **ground-truth bounding boxes** as an alternative supervision signal. Nevertheless, the amodal mask generated by the Bayesian-Amodal approach exhibits **low resolution and uneven boundaries**.
- How to obtain amodal masks with both **high-resolution and accurate boundaries** solely through **box-level supervision**? To deal with this challenge, we propose the **Box-Level supervised Amodal segmentation network through Directed Expansion**, **BLADE**, a weakly-supervised method.

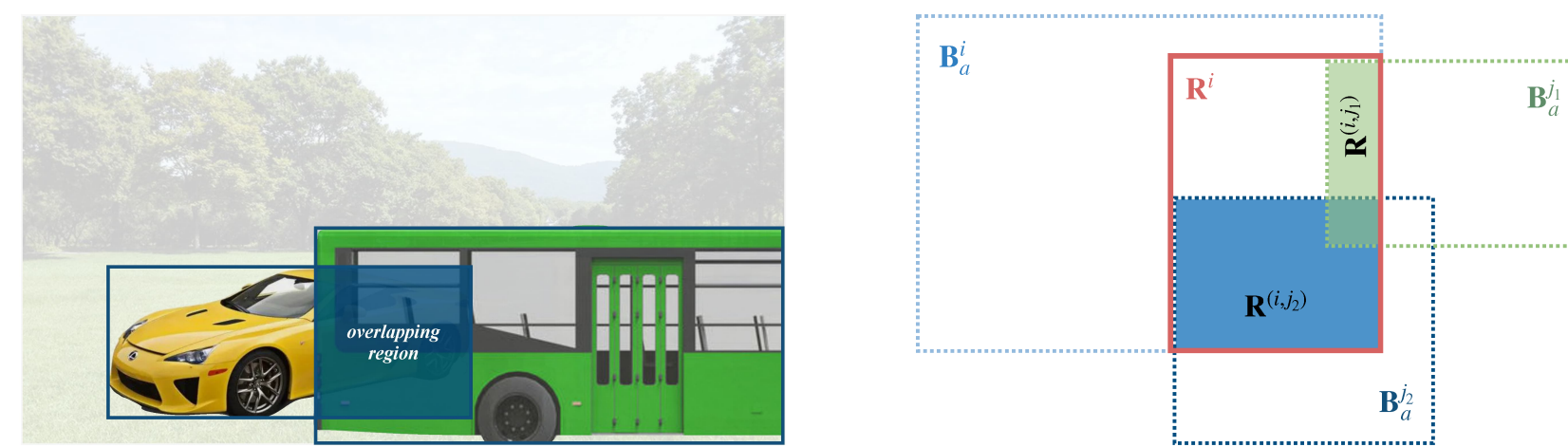
Method

- Network Architectures:** An amodal mask $\mathbf{m}_a$ can be decomposed. Inspired by this, we design a hybrid structure with multiple branches. The three branches share the same multi-scale features extracted from the image and all adopt dynamically-generated instance-aware mask heads containing varying instance-by-instance parameters.
- For the visible-branch, the original mask heads with projection loss and pairwise loss in BoxInst (Tian et al., 2021) are used. $\mathbf{B}_v$ (the bounding box of visible portion) annotations are applied as the supervision. For the amodal-branch, We feed it the predicted $\mathbf{m}_v$ from visible-branch in addition to the features and relative coordinates. $\mathbf{B}_a$ (the bounding box of complete object) annotations are applied as the supervision.

$$\mathbf{m}_a = \mathbf{m}_v + \mathbf{m}_o$$



- Overlapping Region:** In the region-branch, we introduce the overlapping region. The overlapping region of an object is the **tightest** bounding box that covers **all intersecting areas** of its amodal bounding box and those of other objects, so the occluded portion of each object should be inside if exists. If there are multiple intersecting areas, the envelope box is used as the ground-truth overlapping region. For the example in the figure, both $\mathbf{B}_a^{i_1}$ and $\mathbf{B}_a^{i_2}$ overlaps $\mathbf{B}_a^{i_3}$, then the red box $\mathbf{R}^i$ is defined as the overlapping region of instance i .
- We transform the prediction of the four parameters $\mathbf{R}^i = (x_{min}^i, y_{min}^i, x_{max}^i, y_{max}^i)$ of some ground truth bounding box into the prediction of the corresponding **bitmask** to improve robustness while using a simple pixel-level BCE loss.



- Connectivity Loss:** Utilizing the input $\mathbf{m}_v$ as clues, we introduce a **connectivity loss** L_{com} in the amodal-branch to direct the expansion from predicted visible mask $\mathbf{m}_v$ to predicted amodal mask $\mathbf{m}_a$. The overall loss function of amodal-branch is $L^a = \alpha_1^a L_{proj}^a + \alpha_2^a L_{pair}^a + \alpha_3^a L_{con}$.
- The connectivity loss contains two terms, namely neighbor loss and uniform loss. The neighbor loss measures the label consistency of each pixel with its neighbors in $\mathbf{m}_a$, while the uniform loss reflects the consistency of corresponding pixels between $\mathbf{m}_a$ and $\mathbf{m}_v$. The neighbor loss is applied to predicted-overlapping-visible pixels (region ①), while the uniform loss is applied to the whole overlapping region $\mathbf{R}$ (region ①+②).
- Neighbor Loss:** Consider an undirected graph $G = (V_{pov}, E_{pov})$. V_{pov} is the set of predicted-overlapping-visible pixels satisfies $\forall (i, j) \in V_{pov}, (i, j) \in \mathbf{R} \wedge \mathbf{m}_v(i, j) > \tau$ where τ is the thre-

OccludedVehicles

KINS

COCOA-cls



Method	known c	OccludedVehicles									
		FG-0	FG-1	FG-2	FG-3	Mean	FG-0	FG-1	FG-2	FG-3	Mean
BBTP (Hsu et al. 2019)	Yes	66.5	59.7	58.4	57.9	54.4	51.0	48.9	50.4	44.7	53.2
BoxInst (Tian et al. 2021)	Yes	72.3	52.5	53.5	53.9	37.7	38.1	38.2	23.0	22.8	41.6
Bayesian-Amodal (Sun, Kortylewski, and Yuille 2022)	Yes	63.9	59.7	59.6	59.7	57.2	56.8	56.8	55.0	53.9	57.6
Bayesian-Amodal (Sun, Kortylewski, and Yuille 2022)	No	63.0	59.5	59.5	59.5	56.2	55.9	55.6	51.9	50.6	56.0
Ours	No	73.2	70.5	69.7	68.9	69.7	68.1	66.2	68.2	64.5	68.2

Method	Supervision	known c	KINS					COCOA-cls				
			FG-0	FG-1	FG-2	FG-3	Mean	FG-0	FG-1	FG-2	FG-3	Mean
SAM (ViT-H) (Kirillov et al. 2023)	-	Yes	86.7	75.0	50.8	39.0	62.9	82.7	74.9	59.2	42.3	64.8
VRSP (Xiao et al. 2021)	-	Yes	84.7	75.8	74.5	67.1	75.5	82.1	77.7	74.5	72.9	76.8
AISFormer (Tran et al. 2022)	fully	-	85.8	76.4	75.0	69.4	76.7	80.6	76.9	70.9	62.1	72.6
BBTP (Hsu et al. 2019)	weakly	Yes	77.0	68.3	58.9	53.9	64.5	57.3	49.4	40.7	35.0	45.6
BoxInst (Tian et al. 2021)	weakly	Yes	82.0	73.3	56.6	43.6	63.9	76.8	67.0	57.2	34.0	58.8
Bayesian-Amodal (Sun, Kortylewski, and Yuille 2022)	weakly	Yes	72.3	69.6	66.2	58.5	66.7	65.3	65.0	64.3	61.4	64.0
Bayesian-Amodal (Sun, Kortylewski, and Yuille 2022)	weakly	No	69.9	68.1	63.2	47.3	62.1	58.3	59.8	58.6	53.5	57.6
Ours	weakly	No	81.6	74.5	73.7	63.6	73.4	80.3	76.5	69.9	57.9	71.2

shold of the visible-branch, and E_{pov} is the set of edges that connect each pixel with its eight neighbors and contain at least one pixel in V_{pov} . For an edge $e = ((i_1, j_1), (i_2, j_2)) \in E_{pov}$, the ground-truth consistency value $c_e = 1$ when the labels of its two endpoints are the same while $c_e = 0$ when the labels are different.

- The predicted consistency value $\tilde{c}_e$ can be defined as

$$\tilde{c}_e = \mathbf{m}_a(i_1, j_1) \cdot \mathbf{m}_a(i_2, j_2) + (1 - \mathbf{m}_a(i_1, j_1)) \cdot (1 - \mathbf{m}_a(i_2, j_2))$$

which is positively correlated with the consistency and certainty. We adopt the frequently-used BCE loss

$$l_{nc} = -\frac{1}{N_e} \sum_{e \in E_{pov}} c_e \log \tilde{c}_e + (1 - c_e) \log (1 - \tilde{c}_e)$$

to minimize the gap between all $\tilde{c}_e$ and corresponding c_e , where N_e is the number of edges in E_{pov} .

- Uniform Loss:** The values $\mathbf{m}_a(i, j)$ and $\mathbf{m}_v(i, j)$ can be regarded as predictions of the probability that pixel (i, j) belongs to the object and the probability it belongs to the visible portion of the object, respectively. Therefore, it's obvious that any $\mathbf{m}_a(i, j)$ should NOT be less than $\mathbf{m}_v(i, j)$. Observing this, the uniform loss is defined as

$$l_{un} = \frac{K}{N_R} \sum_{(i, j) \in \mathbf{R}} \max(\mathbf{m}_v(i, j) - \mathbf{m}_a(i, j), 0)$$

to penalize those pixels with reduced values from $\mathbf{m}_v$ to $\mathbf{m}_a$, where $\mathbf{R}$ is the set of pixels in the overlapping region and N_R is the number of these pixels.

- By the action of the connectivity loss, an active band is built as the initiation of expansion. Multiple losses for the amodal-branch reach a balance of encouragement and inhibition of expansion thus directing a **moderate** expansion.

Experiments

- Datasets:** Experiments are conducted on three challenging datasets, including **OccludedVehicles** (Wang et al., 2020), **KINS** (Qi et al., 2019) and **COCOA-cls** (Follmann et al., 2019).
- Metric:** We adopt mean Intersection-over-Union (IoU) like the related work Bayesian-Amodal (Sun et al., 2022).
- Baselines:** BBTP (Hsu et al., 2019), BoxInst (Tian et al., 2021), and Bayesian-Amodal (Sun et al., 2022). All these baselines are under box-level supervision, among which BBTP and BoxInst are the two best-performing approaches for box-level supervised general segmentation, and Bayesian-Amodal is a state-of-the-art method for box-level supervised amodal segmentation.
- Input:** Our method only requires the bounding box of the visible portion, which is supported by our expansion-based path. But other methods may require inputting the bounding box of the entire object to know the object center c .
- Comparison:** As shown in the figure and table on the left, the results reflects that our proposed approach **outperforms** existing weakly-supervised methods with large margins and significantly **reduces** the performance gap with those fully-supervised methods.
- Ablation Study:** As shown in the table below, we conduct the ablation study on the KINS dataset, which verifies the effectiveness of the fusion structure, the neighbor loss and the uniform loss. The fusion structure indicates whether to adopt L' and the corresponding region-branch. In addition, small adjustments of the weights in L^a will result in certain but not dramatic performance changes, and our currently selected weights

$$\alpha_1^a = 2.0, \alpha_2^a = 1.0, \alpha_3^a = 1.0$$

achieve good performance.

	UN	NE	FS	FG-0	FG-1	FG-2	FG-3	Mean
1	✓	✓	✓	81.6	74.5	73.7	63.6	73.4
2	✓	✓	✓	81.6	74.9	70.2	57.3	71.0
3	✓	✓	✓	82.8	73.2	56.8	41.6	63.6
4	✓	✓	✓	82.9	72.8	56.7	40.3	63.2
5	✓	✓	✓	76.6	66.7	63.9	56.2	65.9
6	✓	✓	✓	77.3	66.9	62.9	53.2	65.1
7	✓	✓	✓	82.3	74.2	60.0	44.7	65.3
8	✓	✓	✓	82.2	73.1	56.2	40.0	62.9

	α_1^a	α_2^a	α_3^a	FG-0	FG-1	FG-2	FG-3	Mean
1.0	1.0	1.0	1.0	81.3	72.3	69.6	62.1	71.3
2.0	1.0	1.0	1.0	81.6	74.5	73.7	63.6	73.4
1.0	2.0	1.0	1.0	82.0	73.4	72.7	64.8	73.2
1.0	1.0	2.0	1.0	79.9	71.6	68.3	60.1	70.0

